

Y. H. SHANILKA THULSHANI YAPA

BSc. Engineering (Hons)-Computer Engineering (UG), Faculty Of Engineering,University Of Ruhuna

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PROFILE

I am a motivated Computer Engineering undergraduate with a strong interest in software development and emerging technologies. I am confident in working with AI-related concepts and tools, and I bring versatility in adapting to different technologies and problem-solving approaches. I have skills in programming, teamwork, and analytical thinking. I am seeking an internship opportunity to gain hands-on experience and contribute to real-world projects.

TECHNICAL SKILLS

Languages

Python, C, C++, C#, Java, JavaScript, TypeScript, Kotlin, Dart, Go, SQL, Verilog

DevOps & Tools

Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, AWS, CI/CD, Git, Postman, Selenium, Jira

Testing & Databases

Selenium, Postman, TDD, BDD, MongoDB, PostgreSQL, MariaDB, Neo4j

Web & Backend

React.js, Node.js, Express.js, NestJS, MongoDB, PostgreSQL, Prisma, REST APIs, JWT, Firebase

Mobile & Design

Flutter, Figma, Unity, AR Foundation, MATLAB, Proteus

Machine Learning

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, K-Means, PCA, Decision Trees, SVM

ACADEMIC QUALIFICATIONS

Faculty of Engineering, University of Ruhuna, Undergraduate - Computer Engineering 2023 – Present
• CGPA : 3.29

Dhammissara National College 2008 – 2021
• 2018 Ordinary level : 8A's and 1B in English medium
• 2019 Ordinary level (Tamil) : B
• 2021 Advanced level : 2A's and 1B in physical science stream

PROJECTS

ARWays – AR Navigation System, Group Project-Ongoing 2025 – Present
• Developed **ARWayz**, an augmented reality indoor/outdoor navigation system using **Unity + AR Foundation** and **Flutter**, focused on delivering precise, real-time directional guidance. Implemented AR arrow overlays with spatial mapping concepts, QR-assisted indoor navigation workflows, and Google Maps-integrated outdoor routing with live location tracking. Built core mobile features including permission handling, navigation UI flows, and cross-platform Android/iOS support. Currently planning enhancements such as higher-accuracy indoor positioning, dynamic rerouting, and offline navigation capabilities.

Traffic Fine Management System, Full-Stack Group Project 2026
• Developed a full-stack traffic fine management platform with Flutter, React, NestJS, Prisma, and PostgreSQL. Implemented RESTful APIs for user management, traffic fines, payments, and related application workflows. Developed responsive web and mobile interfaces for managing and viewing traffic fines, with backend integration through secure APIs. Integrated payment and notification workflows and used Postman for API testing and validation.

Acadex — Academic Management Mobile Application (Flutter), Individual Project 2026 – Present
• Developed a Flutter-based academic management application for university students to manage GPA, attendance, examinations, assignments, and timetables. Implemented GPA calculation using semester weights, module credits, and grade-point mappings, along with module-wise attendance tracking and percentage calculation. Developed assignment, examination, and timetable management features with reusable UI components and local data persistence using Hive. Designed a consistent and responsive mobile interface using Figma and a custom UI design system.

Book recommendation System 🤖, Machine learning - 2 member group project 2025
• Tech stack includes Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn using google collab. Built a machine learning-based book recommendation system using a Kaggle dataset, including data cleaning, feature engineering, and clustering of books into meaningful reader-focused segments. Implemented and compared K-Means and Agglomerative Clustering with Silhouette Score, Davies-Bouldin Score, and SSE/Elbow analysis, then visualized cluster separation using PCA and distribution plots for interpretability. Developed a cluster-based recommendation workflow for suggesting similar titles and packaged trained artifacts for reuse and deployment integration.

LAN File drop , Group project

- Developed a LAN-based file transfer system with a C++ core enabling fast peer discovery, file sending/receiving, and transfer tracking across local networks. Implemented socket-based networking with UDP broadcast/listening for device discovery, modular sender/receiver components for reliable file delivery, and persistent transfer logging via CSV history, while using CMake-based build configuration to support structured development and deployment on Windows.

Art supply Management system, Database systems - Group project

2025

- Developed a hybrid inventory management system leveraging **MariaDB** for structured transactional data and **Neo4j** to optimize complex supply-chain relationship mapping through graph-based queries.

Qaleesi Volunteering Platform, MERN - Individual project

2025

- Developed a web app using MongoDB, Express, React, Node.js with features like user authentication, profile management, posts (text/images), idea CRUD and messaging.

Software Testing & QA Project, Individual Project

2025

- Developed automated web app tests using TDD, BDD, Selenium, Postman, tracked defects with JIRA, and integrated CI/CD.

FoundIT Lost & Found Mobile App, Group project

2025

- Engineered a mobile application using Kotlin/Flutter to streamline item management; implemented messaging feature using the lost/found item which improved privacy of the users.

LibriCore Library Management System, Devops Individual Project

2025

- Architected and deployed the Libricore full-stack application, reducing manual deployment effort through Infrastructure as Code (Terraform) and automation (Ansible), while implementing a Jenkins-based CI pipeline and a GitHub Actions CD pipeline with Kubernetes on AWS EC2 to enable reliable, scalable, and zero-touch deployments.

EstateEase, GUI Module - Individual project

2024

- Developed a full-stack real estate platform using the MERN stack and SQLite, implementing secure user authentication, an automated loan application workflow, and a persistent cart system, while optimizing data handling and enhancing overall system performance, scalability, and user experience

EstateEase, Desktop app - Individual project

2024

- Developed a **WPF desktop application** featuring a robust **CRUD-based real estate management system** integrated with **SQLite** for high-performance local data persistence and streamlined property tracking.

FitEase, UI/UX Design - Figma individual project

2024

- Designed a cartoon-style exercise app interface in Figma with user-friendly navigation and interactive screens.

CERTIFICATIONS

- Completed the "Python for beginners" by open and distance learning, University of Moratuwa.
- "Python for Everybody" specialization offered by University of Michigan on Coursera.
- Completed "SQL-Basic" and "SQL-Intermediate" in Hackerrank
- Completed the "C# basic" certification on Hacker rank.
- "Excel skills for business" offered by Macquarie University on Coursera.

COMPETITIONS

CodeRally 6.0 organized by IEEE CS Student Branch, IIT

2025

- Competed as a member of Team Nova in the advanced category of the virtual hackathon.

Eminence 5.0 organized by University of Ruhuna

2025

- Placed 8th in a multidisciplinary engineering competition (power systems, ML, network design, innovation pitch) as part of a 5-member team (Team Quadra).

IESL day organized by University of Ruhuna

2024

- Placed 5th in the solid works competition organized by Faculty of Engineering, University of Ruhuna as a part of 2-member team (Team Lightfury)

EXTRA CURRICULAR AND VOLUNTEERING

Rextro Exhibition Team lead and Demonstrator

2025

- Led Finance Management Team, served on Exhibition Committee, and acted as a Demonstrator

Interfaculty Sports Meet (UOR)

2025

- Netball (Champions), Basketball (2nd Runner-up)

Engineering Derby (UOR vs USJ)

2025

- Netball (Champions), Softball Cricket (Champions)

Mehewara Volunteer	2024
• Taught O/L Mathematics to underprivileged students in Puttalam District	
Netwizards Tournament (UOR)	2023 – 2024
• overall champions	
House Captain (School Inter House Sports Meet), Junior prefect	2021

REFERENCES

Dr. Prabath Weerasinghe (B.Sc., M.Sc., Eng., Ph.D.Eng), *Senior Lecturer*, Faculty of Engineering, University of Ruhuna
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